Dynamics of the Stable Boundary Layer: A Geometric Singular Perturbation Framework for Fold Characterization and Regime Transitions

Part 1: Mathematical Foundations and Observational Resolution

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Abstract

Observed critical Richardson thresholds vary widely across field campaigns (e.g., CASES-99 and SHEBA), yielding the long-standing Richardson threshold paradox. We show that this variability is not evidence of incompatible physics, but rather the projection of a multidimensional folded critical manifold onto scalar diagnostics. To formalize this mechanism, we construct a multiscale fast-slow model of the stable boundary layer using Geometric Singular Perturbation Theory (GSPT) on a desingularized coordinate chart. The framework yields a sequence of structural results: existence and smoothness of the critical manifold, fold characterization through fast-Jacobian degeneracy, and a constant-rank projection theorem linking invariant state-space geometry to campaign-dependent observational thresholds. Within this geometry, we define an analytical sensible heat-flux turning-capacity limiter, $H_{\max}$, and a pathway toward dynamic threshold parameterization. Part 1 establishes the mathematical foundations and observational resolution needed for closure development and data-assimilative validation in Parts 2–3.

1 Introduction

1.1 The Crisis of Stability Thresholds in Boundary-Layer Meteorology

1.2 The Closure Trilemma in Atmospheric Modeling

1.3 Objectives and Scope of Part 1

2 Multi-Scale Governing Equations and Chart Regularization

3 Multi-Scale Governing Equations and Chart Regularization

This section defines the smooth fast-slow architecture for the stable boundary layer (SBL) model. We introduce the regularized state space, state the standing assumptions used in the geometric analysis, derive the desingularized fast chart, and define the observation map that connects invariant-manifold structure to atmospheric diagnostics.

3.1 System Definition and Standing Assumptions

We work with a multiscale autonomous system written in the desingularized time variable τ on an open chart domain $\Omega_0 \subset \mathbb{R}^5$:

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{F}(\mathbf{x};\boldsymbol{\mu},\epsilon_1,\epsilon_2) \equiv \begin{pmatrix} F_{\text{fast}}(\mathbf{x};\boldsymbol{\mu}) \\ \epsilon_1 F_{\text{slow}}(\mathbf{x};\boldsymbol{\mu}) \\ \epsilon_1\epsilon_2 F_{\text{superslow}}(\mathbf{x};\boldsymbol{\mu}) \end{pmatrix}. \tag{1}$$

The state vector is

$$\mathbf{x} = \begin{pmatrix} \tilde{e} \\ q_\theta \\ S \\ T_s \\ T_g \end{pmatrix} \in \Omega_0 \subset \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+. \tag{2}$$

Its components are interpreted as follows: $\tilde{e} \in \mathbb{R}_+$ is the desingularized turbulent kinetic energy coordinate, $q_\theta \in \mathbb{R}$ is the kinematic turbulent heat flux $\overline{w'\theta'}$, $S \in \mathbb{R}_+$ is the bulk vertical wind shear $\|\partial\mathbf{u}/\partial z\|$, $T_s \in \mathbb{R}_+$ is the surface skin temperature, and $T_g \in \mathbb{R}_+$ is the deep-soil temperature.

Standing assumptions. Throughout this section we assume:

1. **(A1) Domain regularity.** The chart domain $\Omega_0 \subset \mathbb{R}^5$ is open and connected, and the vector field $\mathbf{F}$ is of class C^∞ on Ω_0.
2. **(A2) Parameter domain.** The parameter block $\boldsymbol{\mu} \in \mathcal{P}$ belongs to an open bounded set of physically admissible coefficients containing the closure, radiative, and thermodynamic constants appearing below.
3. **(A3) Positive invariance.** The physically admissible set

$$\Omega_{\text{phys}} = \{\mathbf{x} \in \Omega_0 \mid \tilde{e} > 0,\ S > 0,\ T_s > 0,\ T_g > 0\} \tag{3}$$

   is positively invariant under the flow.
4. **(A4) Local well-posedness.** For each initial state $\mathbf{x}(0) \in \Omega_0$, there exists a unique local trajectory $\mathbf{x}(\tau)$.
5. **(A5) Timescale separation.** The perturbation parameters satisfy

$$0 < \epsilon_1 \ll 1, \qquad 0 < \epsilon_2 \ll 1, \tag{4}$$

   with fast dynamics on $O(1)$ time, slow dynamics on $O(\epsilon_1^{-1})$ time, and super-slow dynamics on $O((\epsilon_1\epsilon_2)^{-1})$ time.

Unless otherwise stated, we use Ω_0 to denote the full mathematical chart domain and Ω_{phys} to denote the positively invariant physical region used for geometric analysis. We further assume that Ω_0 extends across $\partial\Omega_{\text{phys}}$ to include a neighborhood of $\tilde{e} = 0$, so that $\mathcal{S}_0$ and $\mathcal{C}_{\text{fold}}$ can be analyzed as smooth embedded submanifolds near the laminar boundary.

3.2 Chart Regularization and Desingularization

In the original turbulent kinetic energy variable e, the fast equations contain non-smooth fractional powers through the shear-production term $c_m \ell e^{1/2} S^2$ and the Kolmogorov dissipation term $e^{3/2}/\ell$. As $e \to 0^+$, differentiability is lost at the laminar boundary.

To restore smoothness, introduce the regularized chart map

$$\phi_\delta : e \mapsto \tilde{e} = \sqrt{e + \delta}, \qquad \delta > 0. \tag{5}$$

We formulate the geometric analysis on the regularized chart and use the limit $\delta \to 0^+$ as the asymptotic interpretation of the physical system. To desingularize the fast dynamics, use the positive time reparameterization

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}} \qquad \Longleftrightarrow \qquad \frac{dt}{d\tau} = \epsilon_1 \tilde{e}. \tag{6}$$

Proposition 1 (Regularization and diffeomorphic equivalence)**.** *On the physical domain $e > 0$, the map ϕ_δ is a C^∞ diffeomorphism onto its image. Under the positive time change*

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}}, \tag{7}$$

the fast subsystem extends to a smooth vector field in the chart variable $\tilde{e}$, while preserving phase trajectories, equilibrium sets, and the signs of eigenvalues transverse to normally hyperbolic branches of the critical manifold.

Proof. For $\delta > 0$ and $e > 0$,

$$\phi_\delta(e) = \sqrt{e + \delta} \tag{8}$$

is smooth with derivative

$$\phi_\delta'(e) = \frac{1}{2\sqrt{e + \delta}} > 0, \tag{9}$$

so ϕ_δ is a smooth diffeomorphism onto its image. In the physical variable e, the fast TKE equation is

$$\epsilon_1 \frac{de}{dt} = c_m \ell e^{1/2} S^2 - \frac{g}{\theta_0} q_\theta - \frac{1}{\ell} e^{3/2}. \tag{10}$$

Since $\tilde{e}^2 = e + \delta$, differentiation gives

$$2\tilde{e} \frac{d\tilde{e}}{dt} = \frac{de}{dt}, \tag{11}$$

and hence

$$\frac{d\tilde{e}}{d\tau} = \frac{dt}{d\tau} \frac{d\tilde{e}}{dt} = \epsilon_1 \tilde{e} \cdot \frac{1}{2\tilde{e}} \frac{1}{\epsilon_1} \frac{de}{dt} = \frac{1}{2} \frac{de}{dt}. \tag{12}$$

Substituting the physical equation and using $e = \tilde{e}^2 - \delta$ yields

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} \left(c_m \ell \sqrt{\tilde{e}^2 - \delta}\, S^2 - \frac{g}{\theta_0} q_\theta - \frac{1}{\ell} (\tilde{e}^2 - \delta)^{3/2} \right). \tag{13}$$

In the limiting chart $\delta \to 0^+$ this becomes

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} c_m \ell S^2 \tilde{e} - \frac{g}{2\theta_0} q_\theta - \frac{1}{2\ell} \tilde{e}^3, \tag{14}$$

which is polynomial in $\tilde{e}$. Because $d\tau/dt = \tilde{e}/\epsilon_1 > 0$ on Ω_{phys}, the reparameterization is orientation preserving. Standard results for positive time changes then imply preservation of orbits, equilibria, and the signs of eigenvalues transverse to normally hyperbolic branches. □

3.3 Subsystem Vector Field Components

Under Assumption (A1) and Proposition 2.1, the desingularized vector field components are given explicitly as follows.

Fast subsystem. The fast subsystem $F_{\text{fast}}(\mathbf{x}) = (\tilde{F}(\mathbf{x}), \tilde{H}(\mathbf{x}))^T$ governs rapid turbulent adjustment:

$$F_{\text{fast}}(\mathbf{x}) = \begin{pmatrix} \frac{1}{2} c_m \ell S^2 \tilde{e} - \frac{g}{2\theta_0} q_\theta - \frac{1}{2\ell} \tilde{e}^3 \\ -c_w\, \theta_z(T_s)\, \tilde{e}^3 - \frac{g}{\theta_0} c_\theta \ell q_\theta^2 - \frac{C_\theta}{\ell} \tilde{e}^2 q_\theta \end{pmatrix}. \tag{15}$$

Here the local thermal stratification is represented by the smooth diagnostic map

$$\theta_z(T_s) = \frac{\theta_{\text{top}} - \Pi_s T_s}{\Delta z}. \tag{16}$$

Slow subsystem. The slow subsystem $F_{\text{slow}}(\mathbf{x}) = (F_S(\mathbf{x}), F_{T_s}(\mathbf{x}))^T$ governs wind shear and surface thermal evolution:

$$F_{\text{slow}}(\mathbf{x}) = \begin{pmatrix} \tilde{e}\left[\mathcal{F}_{\text{ls}} - \frac{\partial}{\partial z}\left(c_m \ell \tilde{e} S\right)\right] \\ \frac{\tilde{e}}{C_s}\left[R_{\text{net}}(T_s) + \rho c_p q_\theta + \frac{k_g}{d_g}(T_g - T_s)\right] \end{pmatrix}, \tag{17}$$

where the factors $\tilde{e}$ arise from the positive time reparameterization, so the slow tendencies freeze in fast time as $\tilde{e} \to 0^+$. The corresponding net radiation closure is

$$R_{\text{net}}(T_s) = R_{\text{sw}}^{\downarrow} + \epsilon_a \sigma T_a^4 - \epsilon_s \sigma T_s^4. \tag{18}$$

Super-slow subsystem. The super-slow subsystem governs deep-soil heat conduction:

$$F_{\text{superslow}}(\mathbf{x}) = \tilde{e}\, \frac{k_g}{d_g^2}(T_s - T_g). \tag{19}$$

As in the slow subsystem, the prefactor $\tilde{e}$ is induced by the positive time change; all geometric arguments in subsequent sections are restricted to Ω_{phys}.

3.4 Observation Mapping and Diagnostic Functionals

To connect the internal state $\mathbf{x} \in \Omega_0$ to observable diagnostics, define the smooth observation map

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \to \mathcal{O} \subset \mathbb{R}^3, \tag{20}$$

where $\mathcal{O}$ denotes the space of smooth diagnostic quantities accessible from field observations or numerical model output. Its components are

$$\mathbf{\Pi}_{\text{obs}}(\mathbf{x}) = \begin{pmatrix} \pi_{Ri}(\mathbf{x}) \\ \pi_H(\mathbf{x}) \\ \pi_S(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \frac{g}{\theta_0} \frac{\theta_z(T_s)}{S^2} \\ \rho c_p q_\theta \\ S \end{pmatrix}. \tag{21}$$

These functionals represent the gradient Richardson diagnostic, the sensible heat flux, and the wind-shear coordinate, respectively.

Differential structure of π_{Ri}. The differential covector of the Richardson functional is

$$D\pi_{Ri}(\mathbf{x}) = \begin{pmatrix} 0 & 0 & -\dfrac{2g\,\theta_z(T_s)}{\theta_0 S^3} & \dfrac{g}{\theta_0 S^2}\dfrac{\partial\theta_z}{\partial T_s} & 0 \end{pmatrix}. \tag{22}$$

Hence $D\pi_{Ri}$ lies entirely in the slow cotangent subspace $\operatorname{span}\{dS, dT_s\}$, so the Richardson diagnostic is insensitive, at first order, to perturbations in the fast turbulent directions $(\tilde{e}, q_\theta)$.

Proposition 2 (Observation smoothness). *Under Assumptions (A1) and (A3), the map $\mathbf{\Pi}_{\text{obs}}$ and its component functionals (π_{Ri}, π_H, π_S) are C^∞ on Ω_0.*

Proof. Each component of $\mathbf{\Pi}_{\text{obs}}$ is a composition of smooth algebraic operations on Ω_0. Because $S > 0$ on Ω_{phys}, the denominator in

$$\pi_{Ri}(\mathbf{x}) = \frac{g}{\theta_0}\frac{\theta_z(T_s)}{S^2} \tag{23}$$

never vanishes. The remaining components are linear in q_θ and S, and $\theta_z(T_s)$ is smooth by construction. Therefore $\mathbf{\Pi}_{\text{obs}} \in C^\infty(\Omega_0, \mathcal{O})$. □

Remark 2.2 (Geometry versus observation). The dynamical system evolves on the five-dimensional state manifold Ω_0, whereas atmospheric diagnostics are represented in observation space through the smooth map

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \to \mathcal{O}. \tag{24}$$

Consequently, invariant geometric objects in state space need not appear as invariant objects in observation space.

Section 2 now provides a smooth desingularized dynamical system, an explicit regularized chart, and a diagnostic observation framework suitable for the invariant-manifold results developed in the subsequent sections.

4 Critical Manifold $\mathcal{S}_0$ and Fold Characterization

4.1 Theorem 1: Existence and Smoothness of the Critical Manifold

4.2 Theorem 2: Fold Characterization Theorem

4.3 The Fold Illusion versus Emergent Coupled Catastrophes

5 Projection Theorem and Resolution of the Richardson Paradox

5.1 Diagnostic Projection Mapping π_{Ri}

5.2 Theorem 3: Projection Theorem

5.3 Environmental Constraint Manifolds $\Sigma_{\mathbf{site}}$ and Site Profiles

5.4 Proposition 3.2: Observational Identifiability

5.5 Reconciling Field-Campaign Observations

6 Geometric Taxonomy of Regime Transitions and Nocturnal Life-cycles

6.1 Dimensional-Reduction Hierarchy

6.2 Canonical Singularities and Intermittency

6.3 Four-Phase Nocturnal Relaxation Oscillation

7 Framework for Next-Generation Model Closures

7.1 Analytical $H_{\max}$ Sensible Heat-Flux Limiter

7.2 Dynamic Threshold Parameterization $Ri_{\mathbf{fold}}(T_s, T_g)$

7.3 Adaptive Boundary-Layer Scale Heights $h_{\mathbf{eff}}$

8 Conclusions and Roadmap

8.1 Summary of Theoretical Contributions

8.2 Roadmap for Parts 2 and 3

A Explicit Determinant Derivation of the Fast Jacobian J_f

B Derivation of the $H_{\max}$ Heat-Flux Turning Capacity Formula

C Proof of Transversality for Environmental Constraint Manifolds

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